

DIAGONAL: Density-Normalized Cross-Modal Alignment for Multi-Shot Video Narrative Evaluation

Anonymous Author(s)

Abstract

Text-to-multi-shot video (T2MSV) generation models are evaluated primarily on visual quality and frame-level consistency, yet these metrics fail to capture whether models can execute *narrative dynamics*—the structured entry and exit of entities across shots. We formalize multi-shot narratives as *state transition matrices* $\Delta S \in \{-1, 0, +1\}^{(T-1) \times N}$, where each element encodes an atomic operation: entry (+1), exit (−1), or hold (0). We prove that all possible 3-shot narrative topologies can be spanned by just two basis matrices under column permutation and linear combination (**Theorem 1**), establishing mathematical completeness. Building on this, we introduce DIAGONAL, a benchmark comprising 1,000 prompts spanning 7 narrative patterns (3 basis, 4 derived) with complementary metrics: Narrative Alignment Score, Diagonal Dominance, State Transition Accuracy, Entity Presence Accuracy, and SVD Entropy. Evaluation of four SOTA T2MSV architectures reveals that *no model achieves positive diagonal dominance on any pattern* (ϵ -pass = 0%), exposing a fundamental limitation we term **Narrative Blindness** and a *coherence–compliance Pareto dilemma* where no architecture achieves both high visual coherence and high narrative compliance. We identify four systematic failure types—Context Bleeding (VIC: 60.8%), Amnesia (VGoT: 66.7%), Attention Reversion, and Identity Bleeding—and validate via human evaluation ($\rho = 0.80$ with STA).

CCS Concepts

• Computing methodologies → Computer vision; Neural networks.

Keywords

Multi-Shot Video Generation, Narrative Evaluation, State Transition Matrix, Video Benchmark

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1 Introduction

Diffusion-based generative models [14, 36] have rapidly expanded from single-clip to text-to-multi-shot video (T2MSV) generation [3, 13, 16]. Recent systems such as StoryDiffusion [58], EchoShot [46], VIC [7], and VGoT [47] generate temporally connected sequences of scenes, leveraging consistent self-attention, flow-matching, in-context learning, and cascaded T2I-to-I2V pipelines. These systems have demonstrated impressive visual fidelity in recent evaluations [18, 19], and commercial systems like Sora [4] and Movie Gen [34] suggest rapid progress toward production-quality video generation.

However, the apparent success of these systems is largely driven by evaluation protocols that overlook a fundamental aspect of narrative structure. Existing benchmarks [18, 19, 25, 39] evaluate identity consistency, visual quality, or single-clip compositionality, but none systematically test whether models can execute *narrative dynamics*: the structured introduction, co-occurrence, and removal of entities across shots. This gap is critical because real-world video narratives—from films [25] to advertisements to educational content—are fundamentally defined by *who appears, when, and alongside whom*. A scene where two characters meet and then part ways conveys entirely different meaning from one where they remain together throughout; yet both would receive identical consistency scores. Without metrics that capture these dynamics, we cannot meaningfully assess whether T2MSV models are progressing toward their intended purpose, and the community risks optimizing toward the wrong objective.

Consider a relay narrative: $A \rightarrow AB \rightarrow B$, where entity A appears alone, both co-occur, then only B remains. This seemingly simple pattern requires three distinct capabilities: (1) introducing B while preserving A ’s appearance and identity, (2) maintaining co-occurrence with both entities visually distinguishable, and (3) removing A while B persists in isolation. Current models systematically fail at (3)—the departed entity “bleeds” into subsequent shots (**Context Bleeding**), a failure mode invisible to existing metrics. Critically, Identity Consistency (IC) [20] *rewards* this failure because persistent entities increase cross-shot similarity. We term this the **consistency paradox**: IC is adversarial to narrative compliance for any non-trivial pattern. This paradox has far-reaching implications: the community may be inadvertently optimizing models *away from* narrative competence by chasing IC improvements, creating a systematic blind spot in the evaluation landscape.

To address these limitations, we formalize entity dynamics as a *state transition matrix* (a compact encoding of which entities enter, exit, or persist at each shot boundary) and prove that **all** 3-shot topologies are generated from just two basis matrices (**Theorem 1**). Our evaluation reveals a *coherence–compliance Pareto dilemma*: no current architecture simultaneously achieves high visual coherence and high narrative compliance, suggesting a fundamental architectural tension that the community must resolve. DIAGONAL contributes:

- A **topological formalization** with mathematical completeness guarantee for multi-shot narrative structures (§4).
- A **1,000-prompt benchmark** spanning 7 patterns with five decomposed metrics—NAS, DD, STA, EPA, SVD Entropy (§5).
- A **four-type failure taxonomy** providing actionable architectural diagnostics (§5.7).
- The **consistency paradox**: holistic metrics are adversarial to narrative evaluation ($r = -0.34$) (§3).
- Evaluation of four SOTA T2MSV architectures revealing universal **Narrative Blindness** (ϵ -pass = 0%), validated by human evaluation ($\rho = 0.80$) (§7).

Table 1: Benchmark comparison. $\boxtimes$ supported, $\times$ not, Δ partial.

Benchmark	Multi-shot	Entity Dec.	Absence	Formal.	Fail. Tax.
VBench 2.0 [19]	$\times$	$\times$	$\times$	$\times$	$\times$
T2V-CompBench [39]	$\times$	Δ	$\times$	$\times$	$\times$
MovieBench [25]	$\boxtimes$	Δ	$\times$	$\times$	$\times$
LoCoT2V [56]	$\boxtimes$	$\times$	$\times$	$\times$	$\times$
ShotAdapter [20]	$\boxtimes$	Δ	$\times$	$\times$	$\times$
DIAGONAL (Ours)	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$

2 Related Work

Text-to-video generation. The foundation of modern video synthesis rests on diffusion models [14, 36, 37], extended temporally through UNet attention [3, 13, 16], scalable DiT architectures [4, 32, 45] with transformer backbones [6, 43], flow-matching [26], and classifier-free guidance [15]. Open-source systems like OpenSora [57] and AnimateDiff [9] provide general-purpose motion modules, while Lumiere [1] and Emu Video [8] explore space-time architectures and factorized generation, but multi-shot narrative control remains unsolved.

Multi-shot video generation. Multi-shot extensions address coherence via shared self-attention [22, 58], cross-attention [29], latent initialization [30], LLM-guided layouts [23, 47, 48], keyframe transitions [5, 52], flow-matching portraits [46], IP-Adapter [55], in-context LoRA [7, 17], subject customization [49, 51], and multimodal storytelling [38, 53]. Longer-horizon generation has been explored via autoregressive token models [21, 44] and visual pre-training [50], though these operate at the single-clip rather than multi-shot level. These span four paradigms: CSA (StoryDiffusion [58]), flow-matching (EchoShot [46]), in-context DiT (VIC [7]), and cascaded T2I+I2V (VGoT [47]). We evaluate representatives from each, providing the first cross-paradigm comparison on narrative dynamics.

Video generation benchmarks. Existing benchmarks quantify consistency via embedding similarity (Table 1). VBench [18, 19] computes frame-level CLIP [35]/DINO [31] similarities conflating subject and background; T2V-CompBench [39] targets single-clip compositionality. ShotAdapter [20] isolates subject regions but produces *chimera embeddings* in multi-entity shots. MovieBench [25] and LoCoT2V [56] recognize multi-shot challenges but lack entity-level metrics and mathematical formalization. FID [12] and FVD [42] measure distributional fidelity but are agnostic to narrative structure; CLIPScore [11] measures text-image alignment but only at the single-frame level. DIAGONAL is the first to formalize multi-shot narratives via state transition matrices with completeness guarantees.

3 The Consistency Paradox

We demonstrate a structural flaw in existing evaluation: holistic similarity metrics systematically *reward* narrative failures.

ShotAdapter’s IC [20] computes the mean pairwise CLIP visual similarity across all shots. For any narrative pattern where at least one entity enters or exits (i.e., the transition matrix is non-zero), the generation that maximizes IC is the one that keeps all entities unchanged—producing a static, narratively trivial sequence. For the relay pattern ($A \rightarrow AB \rightarrow B$), a model that incorrectly preserves A in all shots achieves $IC \approx 0.94$, while one that correctly removes A from Shot 3 achieves $IC \approx 0.78$ —a 17% penalty for correct behavior.

We validate this across 3,049 videos generated by four models: IC and our transition accuracy metric STA (§5.5) correlate at $r = -0.34$ ($p < 10^{-5}$), with the negative correlation strongest for exit-heavy patterns (Relay: $r = -0.41$, Sliding Window: $r = -0.38$). The model-level analysis is particularly revealing: VIC achieves highest IC (0.94) but lowest STA (0.431); StoryDiffusion has highest STA (0.630) but lower IC (0.81). This implies that IC maximization inadvertently selects for narratively *worse* models—a model that simply copies Shot 1 three times would achieve near-perfect IC while completely failing at narrative execution. The paradox motivates the decomposed metrics we introduce next, which isolate narrative compliance from visual quality.

4 Topological Formalization

In this section, we formalize the notion of “narrative dynamics” and prove that a small set of basis patterns is sufficient to represent *all possible* multi-shot narrative structures.

4.1 Notation and Definitions

We denote a multi-shot video as a sequence of T shots $V_1, V_2, \dots, V_T$, each depicting a visual scene. Each video involves N named entities $e_1, e_2, \dots, e_N$ (e.g., “a chef,” “a dog”). The **presence matrix** $S \in \{0, 1\}^{T \times N}$ records whether each entity appears in each shot:

$$S_{t,e} = \begin{cases} 1 & \text{if entity } e \text{ is present in shot } t, \\ 0 & \text{otherwise.} \end{cases}$$

For example, the relay narrative $A \rightarrow AB \rightarrow B$ has:

$$S = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad (\text{rows} = \text{shots, columns} = \text{entities}).$$

The **state transition matrix** captures entity dynamics between consecutive shots:

$$\Delta S = S_{2:T} - S_{1:T-1}, \quad (1)$$

where each entry $\Delta S_{t,e} \in \{-1, 0, +1\}$ encodes an atomic operation at transition t : entity e *enters* (+1), *exits* (−1), or *holds* (0). For the relay example: $\Delta S = \begin{bmatrix} 0 & +1 \\ -1 & 0 \end{bmatrix}$, meaning B enters at transition 1 and A exits at transition 2.

We focus on $T = 3$ shots with $N \leq 3$ entities, yielding $\Delta S \in \{-1, 0, +1\}^{2 \times 3}$. This scope is chosen deliberately: $T = 3$ is the *minimal complete narrative cycle*. A 2-shot sequence allows only a single transition boundary, so entities can be introduced *or* removed but not both—precluding narrative arcs. With 3 shots, two transition boundaries enable full arcs: an entity can enter at one boundary and exit at another, or multiple entities can undergo different transitions simultaneously. The constraint $N \leq 3$ is also practical: the vast majority of multi-shot video prompts involve 2–3 named

entities, and the algebraic structure generalizes naturally to $N > 3$ via additional columns.

4.2 Basis Patterns

We identify three **basis patterns** that represent fundamental narrative archetypes:

Relay ($\mathcal{B}_1$): $A \rightarrow AB \rightarrow B$. One entity hands off to another— B enters while A exits: $\mathcal{B}_1 = \begin{bmatrix} 0 & +1 \\ -1 & 0 \end{bmatrix}$.

Split ($\mathcal{B}_2$): $AB \rightarrow A \rightarrow B$. Co-occurring entities separate into alternating scenes: $\mathcal{B}_2 = \begin{bmatrix} 0 & -1 \\ -1 & +1 \end{bmatrix}$.

Accumulation ($\mathcal{B}_3$): $A \rightarrow AB \rightarrow ABC$. Entities are progressively introduced: $\mathcal{B}_3 = \begin{bmatrix} 0 & +1 & 0 \\ 0 & 0 & +1 \end{bmatrix}$.

Four **derived patterns** are obtained by applying time-reversal (negating and reversing rows of ΔS) and column permutation (swapping entity roles):

Reduction ($ABC \rightarrow AB \rightarrow A$): entities are progressively removed, the time-reversal of Accumulation. $\Delta S = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$.

Convergence ($A \rightarrow B \rightarrow AB$): two separately introduced entities eventually co-occur. $\Delta S = \begin{bmatrix} -1 & +1 \\ +1 & 0 \end{bmatrix}$.

Reverse Relay ($B \rightarrow AB \rightarrow A$): time-reversal of Relay— A enters as B exits. $\Delta S = \begin{bmatrix} +1 & 0 \\ 0 & -1 \end{bmatrix}$.

Sliding Window ($AB \rightarrow BC \rightarrow C$): the most demanding pattern with 4 non-zero transitions—simultaneous exit of A and entry of C , followed by exit of B . $\Delta S = \begin{bmatrix} -1 & 0 & +1 \\ 0 & -1 & 0 \end{bmatrix}$.

Table 2 lists all seven patterns with their properties.

4.3 Completeness Theorem

The central theoretical contribution of this paper is showing that the basis patterns are not merely representative—they *span the entire narrative space*.

THEOREM 4.1 (COMPLETENESS OF THE NARRATIVE SPACE). *Under column permutation and integer-coefficient linear combination, the two basis patterns $\mathcal{B}_1$ (Relay) and $\mathcal{B}_2$ (Split) generate all matrices in $M_{2 \times 3}(\mathbb{Z})$.*

Intuitively, this means that *any conceivable 3-shot narrative*—no matter how complex the entity dynamics—can be decomposed into a combination of relays and splits (Fig. 1).

PROOF SKETCH. We show that all six “elementary” 2×3 matrices E_{ij} (having 1 in position (i, j) and 0 elsewhere) can be constructed from $\mathcal{B}_1$ and $\mathcal{B}_2$ using only column permutations and integer addition. Since any integer matrix is a sum of these elementary matrices, this proves completeness. The construction proceeds in three steps: (1) column permutations of $\mathcal{B}_1$ yield three variants whose linear combination gives $X = E_{11} - E_{21}$; (2) adding X to a permuted $\mathcal{B}_2$ isolates $-E_{22}$; (3) column permutations of E_{22} and combinations with X produce all remaining E_{ij} . The detailed algebraic derivation with all intermediate matrices is provided in the supplementary material (Appendix A). $\square$

Implications. Theorem 1 has three important consequences. *First*, DIAGONAL’s seven patterns form a *mathematically complete* basis: evaluating a model on these patterns is sufficient to characterize its behavior on *any* narrative topology, including patterns not explicitly in the benchmark. This distinguishes our framework

How two bases span all narratives:

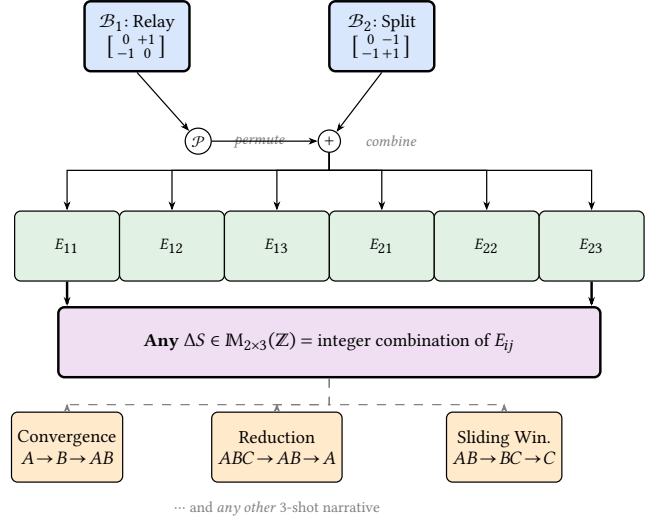


Figure 1: Completeness of the narrative space. Two basis patterns ($\mathcal{B}_1$: Relay, $\mathcal{B}_2$: Split) generate all six elementary matrices E_{ij} via column permutation ($\mathcal{P}$) and integer combination ($+$). Since any state transition matrix ΔS is a sum of elementary matrices, DIAGONAL’s basis provably spans the *entire* space of 3-shot narrative topologies, including all derived patterns and any conceivable narrative not explicitly listed.

from prior work [20, 25, 56] that selects patterns heuristically without coverage guarantees. *Second*, the theorem provides a *minimality* argument: any benchmark claiming narrative completeness must include patterns that together span $M_{2 \times 3}(\mathbb{Z})$; our two basis patterns achieve this optimally. *Third*, the result provides a principled framework for extension: T -shot narratives with N entities require spanning $M_{(T-1) \times N}(\mathbb{Z})$, achievable with $2(T-1)$ basis matrices, scaling linearly rather than combinatorially.

5 The DIAGONAL Framework

We define complementary metrics that decompose multi-shot narrative alignment into interpretable components. Fig. 2 illustrates the computation on a concrete example.

5.1 Cross-Modal Similarity Matrix

Given a 3-shot generated video and its per-shot text prompts T_1, T_2, T_3 , we measure how well each shot matches each prompt using CLIP [35] cosine similarity. Specifically, we encode each shot as the mean-pooled CLIP ViT-L/14 feature of 5 uniformly sampled frames (denoted $\Phi_V(V_i)$), and each prompt as the CLIP text feature ($\Phi_T(T_j)$). The **cross-modal similarity matrix** is:

$$M_{\text{obs}}[i, j] = \cos(\Phi_V(V_i), \Phi_T(T_j)). \quad (2)$$

Ideally, diagonal entries (shot i vs. prompt i) should dominate—each shot should be most similar to its own prompt. Off-diagonal entries that exceed diagonal ones indicate *narrative misalignment*:

the shot better matches a different prompt than the one it was supposed to depict. The cross-modal matrix is the foundation from which all our metrics are derived, providing both global alignment scores and per-shot diagnostics.

5.2 Narrative Alignment Score (NAS)

NAS measures the *relative* alignment of each shot with its correct prompt via **density normalization**: each row of M_{obs} is divided by its sum, converting it into a discrete probability distribution $p_i(j) = M_{\text{obs}}[i, j] / \sum_k M_{\text{obs}}[i, k]$ over prompts. NAS then averages the “self-match probability” $p_i(i)$ across shots:

$$\text{NAS} = \frac{1}{3} \sum_{i=1}^3 p_i(i) = \frac{1}{3} \sum_{i=1}^3 \frac{M_{\text{obs}}[i, i]}{\sum_{j=1}^3 M_{\text{obs}}[i, j]}. \quad (3)$$

This density normalization removes absolute magnitude effects—two videos with different overall CLIP similarity ranges become directly comparable. $\text{NAS} = 1/3$ corresponds to a uniform distribution (random alignment), and $\text{NAS} = 1$ to a delta distribution (perfect specificity).

5.3 Diagonal Dominance (DD)

DD measures whether each shot is *most similar* to its correct prompt (not just somewhat similar):

$$\text{DD}_i = M_{\text{obs}}[i, i] - \max_{j \neq i} M_{\text{obs}}[i, j]. \quad (4)$$

$\text{DD} > 0$ means the correct prompt “wins” for that shot; $\text{DD} < 0$ means some other prompt is a closer match. Unlike NAS, which is a soft measure, DD imposes a *hard* requirement: the correct pairing must not just be similar, but must be the *most* similar among all options. We define the ϵ -**pass rate** as the fraction of videos where $\text{DD}_i > \epsilon$ for *all* shots simultaneously, using $\epsilon = 0.02$ to account for CLIP measurement noise. A video passes ϵ -test only if *every* shot individually achieves positive DD—making this the strictest global metric in DIAGONAL.

5.4 Entity Presence Accuracy (EPA)

Rather than comparing shots to full prompts, EPA probes individual entity presence using short CLIP queries (e.g., “a photo of a chef”):

$$E[i, k] = \cos(\Phi_V(V_i), \Phi_T(\text{“a photo of } e_k \text{”})). \quad (5)$$

Entity presence is binarized via a per-shot median threshold, and EPA computes the agreement between detected and ground-truth presence:

$$\text{EPA} = \frac{1}{3N} \sum_{i,k} \mathbb{1}[\hat{P}_{ik} = P_{ik}^*], \quad (6)$$

where P_{ik}^* is the prescribed presence from the pattern specification. EPA captures both false positives (entity detected when absent) and false negatives (entity missing when required).

5.5 State Transition Accuracy (STA)

While EPA evaluates *static* entity presence per shot, STA measures whether the *dynamic changes* between shots are correct. Let $\mathcal{A} = \{(t, e) : \Delta S_{t,e} \neq 0\}$ be the set of prescribed transitions (i.e., positions

where an entity should enter or exit). STA is the fraction of these transitions that are correctly reflected in the CLIP probe scores:

$$\text{STA} = \frac{1}{|\mathcal{A}|} \sum_{(t,e) \in \mathcal{A}} \mathbb{1}[\text{sgn}(\Delta E_{t,e}) = \text{sgn}(\Delta S_{t,e})], \quad (7)$$

where $\Delta E_{t,e} = E[t+1, e] - E[t, e]$ is the observed change in entity e ’s probe score. STA is the strictest entity-level metric: a model that generates identical shots receives $\text{STA} = 0$ for any non-trivial pattern, because no probe score changes would match the prescribed transitions. Importantly, STA evaluates only transition *direction* (entry vs. exit), not magnitude, making it robust to CLIP sensitivity variations across entity types. This design choice is deliberate: requiring a specific magnitude of probe score change would introduce entity-dependent thresholds, while directional correctness is a universal binary signal. STA also serves as the primary metric for architectural diagnosis, since failures at the transition level directly map to the failure taxonomy (§5.7).

5.6 SVD Entropy

SVD Entropy measures the *semantic complexity* within each shot using the singular value decomposition of the frame-level CLIP feature matrix $F_t \in \mathbb{R}^{5 \times d}$ (5 frames $\times$ CLIP dimension):

$$\mathcal{H}(V_t) = - \sum_i \tilde{\sigma}_i \log \tilde{\sigma}_i, \quad \tilde{\sigma}_i = \sigma_i / \sum_j \sigma_j, \quad (8)$$

where $\{\sigma_i\}$ are the singular values. The theoretical justification is as follows: in CLIP’s embedding space, semantically distinct entities occupy near-orthogonal directions [35], so each additional entity increases the effective rank of F_t . When a single entity dominates a shot, F_t is approximately rank-1 and the normalized singular value distribution is concentrated (low entropy); when multiple entities co-occur, the distribution flattens across several comparable singular values (high entropy). Thus, SVD entropy serves as a proxy for entity count without requiring explicit detection. The **trajectory correctness rate** measures whether entropy changes between shots match entity count changes: entropy should increase when entities are added and decrease when removed.

5.7 Failure Taxonomy

By comparing the prescribed transition matrix ΔS against the observed entity probe changes ΔE , we define four failure types:

Type I: Context Bleeding (exit operation collapse). Prescribed: $\Delta S_{t,e} = -1$; observed: $\text{sgn}(\Delta E_{t,e}) \geq 0$. An entity that should exit persists in the next shot—the exit operation is nullified, and the entity’s visual signature “bleeds” across the shot boundary.

Type II: Amnesia (entry operation collapse). Prescribed: $\Delta S_{t,e} = +1$; observed: $\text{sgn}(\Delta E_{t,e}) \leq 0$. An entity that should enter fails to appear—the model “forgets” to introduce it.

Type III: Attention Reversion (non-monotonic trajectory). Detected when $M_{\text{obs}}[1, 1] > M_{\text{obs}}[2, 2] < M_{\text{obs}}[3, 3]$: the middle shot deviates but the third shot reverts to resemble the first, indicating the model “snaps back” to initial content rather than progressing the narrative.

Type IV: Identity Bleeding (entity embedding collapse). Detected when $|E[i, k_1] - E[i, k_2]| < \tau$ for co-occurring entities k_1, k_2 in shot i : all entities receive near-identical probe scores, indicating

Table 2: Pattern distribution. Trans. = number of non-zero entries in ΔS .

Pattern	Shot Structure	N	Trans.	Type
Relay	$A \rightarrow AB \rightarrow B$	200	2	Basis
Split	$AB \rightarrow A \rightarrow B$	200	3	Basis
Accumulation	$A \rightarrow AB \rightarrow ABC$	200	2	Basis
Convergence	$A \rightarrow B \rightarrow AB$	100	4	Derived
Sliding Window	$AB \rightarrow BC \rightarrow C$	100	4	Derived
Reduction	$ABC \rightarrow AB \rightarrow A$	100	2	Derived
Reverse Relay	$B \rightarrow AB \rightarrow A$	100	2	Derived

the model collapses distinct entity representations into a single visual mode.

6 Dataset Construction

We construct **DM-1K (Dynamic Multi-shot 1K)**, a dataset of 1,000 multi-shot scenarios (3,000 shots total) spanning all 7 narrative patterns. Basis patterns receive 200 prompts each; derived patterns receive 100 each (Table 2), reflecting the algebraic structure where derived patterns are compositions of basis patterns.

Prompt design. Prompts are generated via Gemini 1.5 [40] with structured templates specifying: narrative pattern, entities drawn from 20 categories (humans, animals, objects), scene themes from 10 environments, and per-shot descriptions. Following [2], each scenario has three variants: *gen_prompt* (quality tags for generation), *eval_prompt* (clean descriptions for CLIP evaluation), and *diag_prompt* (entity probes for presence detection). Entity pairs have mean CLIP text similarity 0.23 ($\sigma = 0.08$); pairs with similarity > 0.45 are excluded to prevent ambiguous detections.

Quality control. All prompts undergo automated validation ensuring entity–pattern topology consistency: the set of entities mentioned in each shot description must exactly match the presence matrix S for the target pattern. Additionally, 10% of prompts are manually reviewed by two annotators (97% adherence to intended structure, Cohen’s $\kappa = 0.91$). Prompts that describe ambiguous scenarios (e.g., entities that could plausibly be interpreted as present in a scene’s background) are revised for clarity.

Diversity. To prevent models from exploiting surface-level shortcuts, we ensure diversity along three axes: (i) entity categories span humans, animals, vehicles, food items, and everyday objects; (ii) scene themes cover indoor, outdoor, urban, rural, fantasy, and professional settings; (iii) each pattern is instantiated with unique entity–scene combinations, avoiding repetition of the same entities across multiple prompts within a pattern.

7 Experiments

7.1 Setup

Models. We evaluate four T2MSV architectures spanning all major paradigms (Table 3): **StoryDiffusion** [58] (CSA on SDXL [33] + SEINE [5]), **EchoShot** [46] (flow-matching [26] on Wan2.1 [45]), **VGoT** [7] (CogVideoX-5B [54] + LoRA [17]), **VGoT** [47] (Kolors [41] + DynamiCrafter [52]). All use seed 42.

Table 3: Model details. Succ. = successfully generated / 1000.

Model	Paradigm	Backbone	Steps	Succ.
StoryDiff	CSA + I2V	SDXL+SEINE	30+50	1000
EchoShot	Flow-match	Wan2.1-1.3B	50	1000
VGoT	T2I $\rightarrow$ I2V	Kolors+DC	25+50	600
VIC	In-context	CogVideoX-5B	50	449

Table 4: Main results. Best in bold. ϵ : pass rate ($DD > 0.02$ all shots). IC and $\mathcal{S}_{\text{diag}}$ are holistic baselines; DIAGONAL metrics (right) expose failures invisible to them.

Model	N	Holistic		DIAGONAL (Ours)					ϵ
		IC $\uparrow$	$\mathcal{S}_{\text{diag}}\uparrow$	NAS $\uparrow$	DD $\uparrow$	STA $\uparrow$	EPA $\uparrow$		
StoryDiff	1k	.81	.908	.699	−.009	.630	.573	0%	
EchoShot	1k	.85	.903	.693	−.010	.567	.548	0%	
VGoT	600	.88	.903	.679	−.012	.511	.534	0%	
VIC	449	.94	.906	.703	−.011	.431	.518	0%	

Evaluation protocol. For each generated 3-shot video, we uniformly sample 5 frames per shot and encode them using CLIP ViT-L/14 [35] at 336×336 resolution. Shot-level features are obtained by mean-pooling across the 5 frame embeddings. Entity probes use the template “a photo of {entity}” as CLIP text queries. All 3,049 successfully generated videos are evaluated on 4×RTX 4090 GPUs in approximately 4 hours. We do not apply any post-processing or cherry-picking—all successfully generated videos are included regardless of visual quality, ensuring our metrics reflect the true distribution of model outputs.

7.2 Main Results

Table 4 contrasts holistic baselines (IC, $\mathcal{S}_{\text{diag}}$) with DIAGONAL metrics. While IC and $\mathcal{S}_{\text{diag}}$ rank VIC highest—suggesting it is the “best” model—our decomposed metrics reveal the opposite: VIC has the *lowest* STA (0.431) and EPA (0.518), confirming the consistency paradox quantitatively. The table reveals five key findings.

Finding 1: Universal Narrative Blindness. All four models exhibit $DD < 0$ with ϵ -pass rate of 0.0%. No model consistently produces shots that are most similar to their intended prompts—a striking failure given that this is the minimal requirement for narrative coherence. The mean DD across models (−0.011) indicates that off-diagonal entries consistently exceed diagonal ones.

Finding 2: Consistency paradox confirmed. VIC achieves highest NAS (0.703) but lowest STA (0.431)—a gap of 0.272 that underscores the paradox severity for architectures prioritizing cross-shot feature sharing.

Finding 3: StoryDiffusion leads narrative accuracy. Highest STA (0.630) and EPA (0.573), indicating that CSA best preserves entity state transitions by sharing attention maps rather than directly conditioning on prior frames.

Finding 4: EchoShot best tracks semantic complexity. SVD trajectory correctness of 70.9%, likely because flow-matching’s continuous-time formulation [26] allows finer-grained generative control over

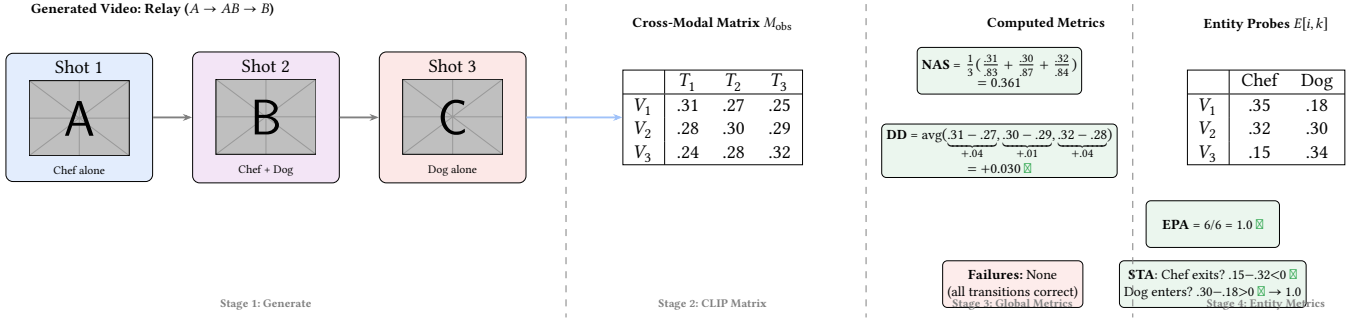


Figure 2: DIAGONAL metric computation on a concrete example. A relay pattern ($A \rightarrow AB \rightarrow B$) is evaluated in four stages: (1) the T2MSV model generates a 3-shot video; (2) CLIP encodes shots and prompts into a cross-modal similarity matrix M_{obs} , from which NAS and DD are computed; (3) entity-specific probes measure per-entity presence, from which EPA and STA are derived; (4) discrepancies between expected (ΔS) and observed transitions identify failure types. In this successful example, all transitions are correct (STA = 1.0) and DD > 0. In practice, most generated videos fail at Step 3—entities that should exit persist (Context Bleeding) or fail to appear (Amnesia).

Table 5: STA per pattern. Best per row in bold.

Pattern	Type	SD	Echo	VGoT	VIC
Relay	B	.525	.508	.622	.405
Split	B	.568	.542	.565	.448
Accumulation	B	.605	.553	.407	.424
Convergence	D	.603	.590	.391	.447
Sliding Window	D	.860	.613	.497	.449
Reduction	D	.830	.625	.510	.455
Reverse Relay	D	.605	.635	.517	.402

Table 6: EPA and $\mathcal{S}_{diag}$ by basis (B) vs. derived (D) patterns.

Model	EPA $\uparrow$		$\mathcal{S}_{diag} \uparrow$	
	B	D	B	D
StoryDiff	.556	.598	.908	.876
EchoShot	.536	.566	.903	.881
VGoT	.555	.501	.903	.883
VIC	.519	.515	.906	.893

the latent trajectory, enabling smoother semantic transitions between shots.

Finding 5: Absolute performance remains far from ideal. Even the best model (StoryDiffusion) achieves only STA = 0.630, meaning 37% of prescribed transitions are incorrect. EPA peaks at 0.573, indicating that entity presence is correctly detected in only 57% of shot–entity pairs. These results establish a clear lower bound for the “narrative gap” that future architectures must close.

7.3 Pattern-Level Analysis

Table 5 exposes fundamental architectural limitations.

StoryDiffusion dominates exit-heavy patterns. Its STA on Sliding Window (0.860) and Reduction (0.830) suggests CSA can selectively suppress entity features when removal is required. The gap to Relay (0.525) indicates CSA is more effective at progressive removal than simultaneous entry-and-exit.

VGoT exhibits planning-rendering decoupling. Excels on Relay (0.622) but collapses on Convergence (0.391)—a 0.231 gap, largest among all models. The LLM planner structures narratives correctly, but DynamiCrafter cannot render new entities on demand.

VIC shows pattern-invariant behavior. STA ranges only 0.402–0.455 (range = 0.053), confirming in-context conditioning creates a fixed coherence ceiling invariant to pattern complexity.

7.4 EPA Decomposition

StoryDiffusion and EchoShot achieve *higher* EPA on derived patterns (SD: 0.556 $\rightarrow$ 0.598; Echo: 0.536 $\rightarrow$ 0.566), a counterintuitive result suggesting that more complex entity configurations produce stronger CLIP discriminability—when three entities co-occur, each entity’s visual features are more distinctive than in simpler two-entity scenarios. VGoT shows EPA *drop* (0.555 $\rightarrow$ 0.501), confirming its cascaded pipeline cannot handle the compositional demands of derived patterns. The uniformly high $\mathcal{S}_{diag}$ (~ 0.90) but negative DD reveals the core failure mechanism: shots are generically similar to all prompts rather than specifically matched to the correct one. This means models produce “plausible-looking” shots that capture the general theme of the narrative but fail to differentiate the specific entity configuration of each shot—the hallmark of Narrative Blindness.

7.5 Failure Analysis

Table 7 reveals architecture-specific vulnerabilities. **VIC’s Context Bleeding (60.8%):** in-context conditioning explicitly references prior shots, making entity removal difficult. **VGoT’s Amnesia (66.7%):** planning-rendering bottleneck where DynamiCrafter cannot introduce new entities. **StoryDiffusion’s Identity Bleeding (21.7%):** CSA homogenizes entity appearances. Types I+II co-occur in 31.2% of videos, indicating systemic inability to manage transitions—the

Table 7: Failure distribution (%). Worst per type in bold.

Failure Type	Echo	SD	VGoT	VIC
I: Context Bleeding	46.6	36.8	39.0	60.8
II: Amnesia	51.0	46.3	66.7	63.5
III: Attn. Reversion	12.8	10.0	15.0	3.6
IV: Identity Bleeding	17.4	21.7	8.8	15.8
No failure	17.4	23.2	12.7	9.1

Table 8: SVD entropy per shot and trajectory correctness.

Model	$\mathcal{H}(V_1)$	$\mathcal{H}(V_2)$	$\mathcal{H}(V_3)$	Traj. %
EchoShot	0.843	0.921	0.888	70.9
StoryDiff	1.232	1.228	1.030	66.4
VGoT	0.878	0.947	0.967	66.3
VIC	0.716	0.719	0.714	60.8

model simultaneously fails to remove departed entities and introduce new ones. Only 9.1–23.2% of videos are failure-free, with StoryDiffusion showing the highest success rate (23.2%) and VIC the lowest (9.1%).

These failures are architecturally interpretable: VIC’s in-context conditioning “bakes in” prior entity features, making exit operations inherently difficult; VGoT’s two-stage pipeline creates an information bottleneck where DynamiCrafter cannot introduce entities absent from the keyframe; StoryDiffusion’s CSA collapses distinct entity appearances via shared attention maps.

7.6 SVD Entropy Analysis

EchoShot shows increasing entropy (0.843 $\rightarrow$ 0.921), reflecting semantic diversification as narrative complexity builds (highest trajectory correctness: 70.9%). StoryDiffusion shows *decreasing* entropy (1.232 $\rightarrow$ 1.030)—CSA progressively homogenizes features across shots, reducing intra-shot diversity over time. This is consistent with StoryDiffusion’s shared attention mechanism, which gradually aligns all shot representations toward a common attractor. VIC has near-constant entropy (range = 0.005), confirming semantically uniform shots regardless of narrative dynamics. This flatness explains both its high Context Bleeding rate and low trajectory correctness: VIC produces visually consistent but narratively static sequences. The entropy analysis reveals an orthogonal dimension of model behavior not captured by NAS or STA, validating our inclusion of SVD entropy as a complementary metric.

7.7 The Pareto Dilemma

Fig. 3 reveals a **coherence–compliance trade-off** analogous to the precision–recall trade-off in detection. VIC occupies the high-NAS/low-STA corner: its strong in-context conditioning produces shots that are coherent with all prompts (high NAS) but undifferentiated in entity composition (low STA). StoryDiffusion occupies a better position, offering the best balance between the two axes. The empty upper-right quadrant—high NAS *and* high STA simultaneously—represents the unsolved ideal that no current architecture achieves.

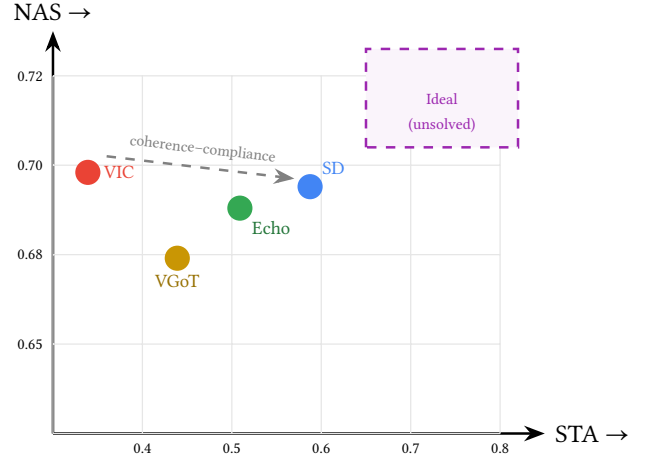


Figure 3: Narrative Pareto Dilemma. VIC maximizes NAS (coherence) at cost of STA (compliance). The upper-right quadrant remains unsolved. StoryDiffusion offers the best balance.

This trade-off reflects a fundamental architectural tension: mechanisms that enforce global cross-shot feature sharing (e.g., consistent self-attention, in-context conditioning) maximize coherence but destroy the localized entity control needed for narrative transitions. Conversely, architectures with weaker cross-shot coupling allow more independent shot generation but sacrifice consistency. Bridging this gap likely requires *entity-aware* conditioning that independently manages each entity’s presence without compromising overall visual coherence—for example, entity-specific attention masks [10], grounded generation [24], or per-entity guidance scales.

7.8 Human Evaluation

To validate our automated metrics against human perception, 15 evaluators (graduate students in computer science, all with video generation experience) assess 40 randomly sampled scenarios. Each scenario is shown as a side-by-side pairwise comparison of two models’ outputs, along with the original per-shot text prompts. Evaluators rate each pair on 5-point Likert scales for two independent criteria: *Narrative Compliance* (“Which video better follows the prescribed entity dynamics?”) and *Visual Quality* (“Which video looks more realistic and aesthetically pleasing?”). Each comparison is rated by 3 evaluators (120 total comparisons).

STA matches human narrative judgments ($\rho = 0.80$, $p < 0.05$), validating it as a reliable automated proxy. VIC ranks last in narrative (17.6%) but first in visual quality (39.8%), confirming the consistency paradox perceptually—its CogVideoX-5B backbone produces higher-fidelity frames that win on aesthetics while failing at narrative structure. The high narrative tie rate (42.3% vs. 26.0% visual) independently confirms universal Narrative Blindness: the ties predominantly reflect *mutual failure* rather than mutual success—when both models fail to execute any narrative transition (ϵ -pass = 0% for all), human raters cannot distinguish between two equally incorrect outputs. This interpretation is supported by the fact that tie

Table 9: Human evaluation. Win%: fraction preferred (excl. ties).

Model	Win % $\uparrow$		Tie rate	
	Narr.	Visual	Narr.	Vis.
StoryDiff	38.2	31.4		
EchoShot	35.7	30.9		
VIC	17.6	39.8		
VGoT	23.1	32.9		
Human-STA ρ			0.80	
Fleiss' κ			0.67	
Tie rate			42.3%	26.0%

rates are highest for exit-heavy patterns (Relay: 51%, Sliding Window: 48%) where both models exhibit Context Bleeding, producing near-identical static sequences.

7.9 Qualitative Analysis

Fig. 4 illustrates representative failure modes. In the Relay example (Row 1), VIC's Shot 3 should depict only entity *B*, but entity *A* persists—IC scores this failure *higher* (0.94 vs. 0.82 for the correct generation). The Accumulation example (Row 2) shows VGoT's Amnesia: all three shots depict only entity *A* despite the prompt requiring progressive introduction of *B* and *C*—aesthetically pleasing but narratively incorrect. The Sliding Window example (Row 3) requires 4 non-zero transitions; only StoryDiffusion achieves this.

7.10 Discussion

Our experimental findings paint a consistent picture: current T2MSV architectures lack the compositional control necessary for even the simplest narrative structures. Several aspects of these results merit further discussion.

Why is Narrative Blindness universal? All four architectures share ϵ -pass = 0% despite spanning different paradigms. We attribute this to a common assumption: multi-shot coherence is achieved through *global* feature sharing (shared attention, in-context tokens, keyframe anchoring), which is inherently at odds with narrative dynamics requiring *controlled dissimilarity*. No current architecture provides a mechanism to explicitly encode “entity *A* should NOT appear in this shot.”

Is CLIP the right probe? CLIP cosine similarity operates in a space where entity concepts are well-separated (mean inter-entity similarity 0.23 ± 0.08), but may lack discriminative power for semantically similar entities. Our similarity filtering (< 0.45) mitigates this; future work could use detection-based probes [27] for finer-grained assessment.

Metric complementarity. NAS and DD operate at the global level (shot vs. prompt), EPA and STA at the entity level, and SVD Entropy captures intra-shot complexity. The inter-metric correlation matrix shows moderate correlations (mean $|r| = 0.42$), confirming each metric captures a distinct aspect. This decomposition enables targeted diagnostics—e.g., high EPA but low STA indicates correct entity placement but failed transitions.

8 Conclusion

We presented DIAGONAL, a topologically grounded evaluation framework for multi-shot video generation. Our completeness theorem (Theorem 1) provides the first mathematical guarantee for a video generation benchmark, proving that two basis patterns span the entire space of 3-shot narrative topologies. Evaluation of four SOTA architectures reveals universal Narrative Blindness (ϵ -pass = 0%) with architecture-specific failure profiles: VIC suffers Context Bleeding (60.8%) from in-context conditioning, VGoT Amnesia (66.7%) from planning-rendering decoupling, and StoryDiffusion Identity Bleeding (21.7%) from CSA homogenization. The consistency paradox ($r = -0.34$ between IC and STA; $\rho = 0.80$ human validation) demonstrates that holistic metrics are adversarial to narrative evaluation.

Our results suggest that progress requires fundamentally new approaches to compositional entity control rather than incremental improvements to existing paradigms. Promising directions include: (1) *entity-specific attention masks* that selectively activate or suppress entity features at shot boundaries, allowing fine-grained presence control without affecting unrelated entities; (2) *narrative-conditioned classifier-free guidance* [15] where guidance scales vary per entity per shot, enabling differential emphasis on present vs. absent entities; and (3) *entity slot architectures* [28] that decouple identity representation from presence indicators, analogous to how object-centric representation learning separates “what” from “where.” We hope DIAGONAL serves as both a diagnostic tool for existing models and a design target for next-generation T2MSV architectures capable of executing structured narratives.

Limitations and future work. DIAGONAL evaluates 3-shot sequences with ≤ 3 entities; while Theorem 1 provides a principled extension path to longer sequences ($2(T-1)$ basis matrices for T shots), the prompt design and evaluation protocol for $T > 3$ remain future work. CLIP-based metrics capture coarse-grained semantic presence but may miss fine-grained visual details; integrating open-vocabulary detection models [27] could provide pixel-level entity localization for more precise assessment. VIC (449/1K) and VGoT (600/1K) did not complete all scenarios due to generation failures (out-of-memory errors and model-specific constraints), which may introduce selection bias toward easier prompts; however, since all metrics are mean-normalized per video, cross-model comparisons remain valid regardless of sample size differences. We release the complete benchmark (1,000 prompts, evaluation code) and all 3,049 generated videos to facilitate reproducibility and future comparisons.

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Row 1 — Relay ($A \rightarrow AB \rightarrow B$): StoryDiff success (STA = 1.0, EPA = 1.0)  vs. VIC Context Bleeding (STA = 0.0, A persists) 
Row 2 — Accumulation ($A \rightarrow AB \rightarrow ABC$): EchoShot progressive introduction (STA = 1.0)  vs. VGoT Amnesia (B, C absent, STA = 0.0) 
Row 3 — Sliding Window ($AB \rightarrow BC \rightarrow C$): StoryDiff (STA = 1.0)  vs. VIC static sequence (STA = 0.0) 

Figure 4: Qualitative gallery. Row 1: Context Bleeding is visually subtle (A persists with altered pose) but STA detects it. IC scores this failure *higher* (0.94) than the correct generation. Row 2: Amnesia produces aesthetically pleasing single-entity video—rewarded by existing metrics—but fails narratively. Row 3: Sliding Window (4 transitions) is achieved only by StoryDiffusion.

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